

Yves Mojica

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Education

University of California, San Diego

September 2022 - June 2026

– Bachelor of Science in Computer Engineering, GPA: 3.27

Experience

Hermes Avionics Engineer | Rocket Propulsion Lab

September 2023 - June 2026

- Develop code for various IMU sensors on Arduino and ESP32 microcontrollers in C++ for flight data collection of altitude, temperature, location, and acceleration of our rocket during flight
- 3D printed and designed avionics system parts to transform concepts into electrical components onboard our rocket, as well as soldering all our electrical components for a sturdy and secure avionics bay
- Conduct hands-on circuit testing/troubleshooting for avionics systems using lab equipment to debug issues

Sales Advisor | H&M

August 2025 - Present

- Adapt and solve problems for customers in a fast-paced environment to provide a positive experience for shoppers which improved sales and efficiency for our store

Vice President - Finance | Rocket Propulsion Lab

April 2025 - June 2026

- Organize and manage bill of materials for all of our projects at RPL with spreadsheets, allowing for easier tracking of orders for myself and future administration members
- Communicate and facilitate meetings with team leads and advisors about budgets or other financial concerns to accurately track finances

Projects

New Member Project | Rocket Propulsion Lab

October 2022 - May 2023

- Managed and led a team of 5 in a new member project to design a small-scale rocket using, 3D printed parts, and an avionics system on an Arduino nano that successfully launched
- Designed and circuit tested an avionics bay in SolidWorks to house an Arduino nano, altimeter, and a battery that could slide inside our rocket and record basic flight data for post flight analysis

Radio Telemetry Tracking | Engineers For Exploration

November 2024 - June 2025

- Collaborated in a research project team to develop technologies for local organizations to aid in research efforts
- Sketched and designed an updated CAD with SolidWorks for our drone payload to house new electrical components and manage wiring for an improved updated design

Automated Robocar/Jeepbot

January 2026 - March 2026

- Built a 1/8th scale robot car that could navigate tracks fully automated using a Raspberry PI for smart control
- Wired, soldered, and crimped cables for hardware components of our robocar/jeepbot, as well as using CAN communication protocol for coding our speed controllers in Python
- Designed and transferred this hardware onto a kid's jeep, where we replaced the stock electronics with microcomputers, managed cabling, and custom wiring throughout for large scale automated navigation

Qualcomm Cluster

March 2026 - Present

- Design and improve upon an existing server cluster project sponsored by Qualcomm, by providing a proposal and bill of materials that vastly reduced costs and made reproducible documentation for future development
- Managed a project team to design a new power system consisting of a custom PCB using KiCad, the testing of Rubik Pi 3s under non traditional power draw, and benchmarking of a multi-node server cluster operating on Kubernetes

Skills

Programming: C, C++, Java, Python, MATLAB, HTML/CSS, SystemVerilog

Technical: LTspice, 3D printing, Arduino/ESP, Raspberry Pi, CAD Onshape, Solidworks, Git, KiCad, Microsoft Office

Equipment: Function generators, Oscilloscopes, Multimeters, Soldering irons, Power Supplies